

IT362: Principles of Data Science

The Changing Landscape of Programming Languages

Logbook

Prepared By

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Phase	Task	Date	Description	Challenges
Phase 1: Data Collection, Research, and Assessment	Accessing GitHub API to collect data	September 18 th	We started using python in google colab to apply the collecting code. We decided to collect 1000 observations for each year (250 per quarter) except for 2025 we ended up collecting almost 750 observations. The features we decided to collect for each repository were ["name", "owner", "url", "description", "stars", "forks", "created_at", "pushed_at", "primary_language", "all_languages_bytes", "topics", "contributors_count"] to expand on the study.	API Rate Limits & Time-consuming collection, we solved this issues by dividing the task on the group members to get it done faster.
	Finalizing the raw data file	September 23 rd	We gathered all the data collected by the group in one file and made sure they are compatible.	Merging all years' files together smoothly, we solved this issue by merging them via python then checking manually.
	Data checking and visualizing.	September 24 th	We used python libraries such as numpy and matplotlib and seaborn to: -Check for any duplicate rows or missing values, there were no duplicates, but we found some missing values and we decided to only remove rows with missing "Primary programming language" value.	Hesitation regarding wither to remove all rows with any missing values or only the ones that misses "Primary programming language", eventually we decided to only remove rows with missing "Primary programming language" value only.

	Writing the 1 st phase's report	September 22 nd	We started writing the 1 st draft for this phase's report.	No challenges faced.
Phase 2: Data Processing, Cleaning, and Exploratory Data Analysis (EDA)	Checking for dataset size	October 12 th	We used statistically sufficient for analysis by calculating the required sample size to ensure the dataset has enough observations for the analysis. Applied the sample size formula $n = \frac{Z^2 * p * (1-p)}{e^2}$ using a confidence level of 95% and a 5% margin of error.	No challenges faced. The dataset was found to be sufficient for analysis, so we proceeded with it.
	Numerical Feature Transformation	October 13 th	Applied Logarithmic Transformation ($\log(1+x)$) to the stars and forks columns. This was crucial for handling the highly skewed distributions and mitigating the impact of extreme outliers, preparing the data for robust statistical modeling.	The main challenge was selecting the appropriate transformation technique to normalize the non-normal distributions of popularity metrics, which was addressed effectively by using the $\log(1+x)$ function
	Text and Temporal Feature Engineering	October 13 th	Handled remaining missing values in text columns (description and topics) by replacing them with empty strings (' ') to allow for subsequent Text Vectorization. - Calculated the repository age in days (repo_age_days) from the 'created_at' timestamp for temporal analysis.	Ensuring that the missing text values were handled in a way that preserves the data structure
	Temporal Trend Analysis (Repository Creation)	October 13 th	Analyzing the "Repository Creation Trend Over Time" graph. The analysis showed an increasing trend in the number of AI repository creations since 2020, with a noticeable rise in activity towards the end of 2024 and	Ensuring the date column was correctly formatted and grouped by month (<code>.dt.to_period("M")</code>) to accurately

			early 2025. This confirms the growing interest and activity in the AI/ML domain over the past few years.	plot the monthly trend was a critical step in this analysis.
	Temporal Trend Analysis (Average Popularity)	October 14 th	Analyzed the "Average Stars per Month (2020-2025)" graph. The visualization shows a significant, sporadic increase in the average number of stars obtained by new repositories, particularly noting a sharp spike in late 2024/early 2025. This suggests that while more projects are being created, the average popularity of new projects has also seen periods of major growth.	The key challenge was interpreting the meaning of the spikes (which represent one or two highly starred projects in that month) versus the general baseline trend.
	Advanced EDA: Language Analysis Refinement	October 14 th	- Refined the primary language analysis by merging 'Jupyter Notebook' entries with 'Python' to accurately represent the core programming language in AI projects. - Identified the most frequent primary languages and common co-occurring secondary languages (e.g., Shell, JavaScript) to understand project technology stacks.	The classification needed refinement to treat 'Jupyter Notebook' as an environment for 'Python' code rather than a standalone language.
	Advanced EDA: Topic & Correlation Analysis	October 14 th	Analyzed the Top 10 most common topics associated with the AI/ML repositories (e.g., machine-learning, deep-learning, llm, openai). - Computed the correlation coefficient between 'repo_age_days' and 'stars_log', concluding that age has a very weak correlation (0.028) with popularity.	The primary challenge was extracting meaningful insights from the repository topics, which was solved by focusing on high-frequency keyword analysis.

	Writing the 2 nd phase's report	October 15 th	We started writing the 2 nd draft for this phase's report.	No challenges faced.
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